Animesh Mishra

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Education

- **New York University(NYU) - Courant Institute of Mathematical Sciences** New York, USA
  Masters of Science - Computer Science (Concentration in AI) *September 2023 - May 2025*
  Courses: *Deep Learning Systems, Operating Systems, Programming Languages, Computer Vision, Big Data and ML Systems, Natural Language Processing*
- **Birla Institute of Technology, Mesra** Ranchi, India
  Bachelor of Technology - Computer Science and Engineering; GPA: 8.64 *August 2019 - May 2023*
  Courses: *Machine Learning, Advanced Data Structures and Algorithms, Computer Networks, Database Management Systems*

Experience

- **NYU Langone Health** New York, USA
  Machine Learning Engineer *January 2024 - Present*
  - Built large-scale **ETL pipelines** (scaling to terabytes) with **Databricks, Spark, Kubeflow, AWS S3**, and **PostgreSQL** to capture and process fMRI readings and patient behaviors. Orchestrated **Kubernetes** clusters on **AWS EKS** to run **multimodal LLMs** for real-time data capture and preprocessing, achieving a **10x** reduction in preprocessing time.
  - Fine-tuned transformer-based models in **PyTorch** and **LoRA Adapters** to predict patient behaviors with **95.3% accuracy**. Optimized **HPC** cluster throughput by **300%** through **Distributed Training, NCCL, Prometheus, Grafana, OpenMPI**, and **CUDA**.
  - Automated **data collection** and **testing** with **Selenium** and **AWS Lambda**, cutting manual effort by **3x**. Hosted a **ReactJS** + **Redux** dashboard with **Django REST APIs** for real-time visualization.
  - Established robust **CI/CD** pipelines with **Jenkins** to automate builds, testing, and deployments in an **Agile** environment, ensuring high code quality and reducing integration overhead.
- **Microsoft** Bengaluru, India
  Software Engineering Intern *May 2022 - July 2022*
  - Developed REST APIs for the Azure Storage **Hadoop Distributed File System (HDFS)**, **Blob Storage** and **Data Lake Storage Gen1** endpoints using **C#** and **.NET** resulting in a **20% increase** in service efficiency.
  - Added multiple Data Integrity checks to Data Clusters within a **microservices** framework using **SHA-256** and **MD5** encryption schemes, significantly reducing negligence in data corruption by **over 60%**.
  - Enhanced internal tools for **load testing**, **unit testing**, **cloud monitoring**, and **CI/CD** pipelines within Azure Storage, improving system reliability by over 15% within an **Agile** development environment.
- **Tata Steel** Jamshedpur, India
  Summer Trainee *June 2020 - August 2020*
  - Built ML models based on **LSTM**- and **XGBoost** to predict **welding quality** in **Cold Rolling Mills**, reducing defects by **20%**.
  - Efficiently processed large **time-series** datasets using **Fourier transforms**, **PCA** for preprocessing, and a novel **LSTM** based approach, achieving over **98.7% accuracy**.
  - Improved **process efficiency** by **35%**. Deployed an **Angular**-based UI for real-time monitoring, integrating **PostgreSQL** for data management and **Kafka** for streaming process status.

Projects

- **Vartalaap (Full Stack development)**: Engineered a communication application utilizing **ReactJS**, **Redux**, **RabbitMQ**, **Node.js**, **Socket.IO**, **MongoDB** and **WebRTC**, incorporating **OAuth**-based user authentication, multiple-peer video conferencing, and chat functionalities. Link
- **Stock.ai (AI, Finance)**: Designed a scalable **RAG**-based stock prediction system using **GPT**, **LangChain**, and advanced prompt engineering. Deployed on **GCP**, integrating AI-driven insights into a recommendation engine for informed financial decisions. Link
- **StrokePlayAR (Augmented Reality, TypeScript, LLM)**: Created an AR-based rehabilitation game for stroke survivors using **Lens Studio**, **OpenGL**, **TypeScript**, and **Metal**. Implemented gesture recognition with real-time feedback and adaptive difficulty for motor skill recovery. Integrated an **LLM**-powered **RAG** system to enhance rehabilitation outcomes. Link
- **CEMDL (CUDA, C++, GPU Programming)**: Designed **CUDA kernels** for auto-differentiation, **GEMM** operations, and architectures including **ResNet-18**, **attention** modules, and **Transformers**, enhancing performance and scalability. Optimized **memory management** and leveraged **CUTLASS** and **CUDA streams** for efficient parallel processing in machine learning models. Link

Publications

- **A. Mishra**, R. Jha and V. Bhattacharya, "SSCLNet: A Self-Supervised Contrastive Loss-Based Pre-Trained Network for Brain MRI Classification", **IEEE Access** [DOI]
- **A. Mishra** and V. Bhattacharya, "Applying Semi-Supervised learning on Human Activity Recognition Data", **ICIBT** [DOI]

Skills

- **Languages**: Python, C/C++, JavaScript, C#, SQL, Java, .NET
- **Skills**: PyTorch, TensorFlow, CUDA, Keras, Kubeflow, Hadoop, HDFS, Node.js, CMake, React, Typescript, Redux, Django, Kubernetes, Docker, GIT, MySQL, MongoDB, Linux, Windows, Azure Files, Amazon Web Services(AWS), Google Cloud Platform (GCP), JSON, XML, Firebase, Swagger API, OpenCL, Deep Graph Library (DGL), Wordnet, PostgreSQL, CUTLASS, Natural Language Processing, Web-development, Machine learning, High Performance Computing, GPU Programming, Software Development Life Cycle (SDLC), MapReduce, Unit Test,Postgres, Regression, Version Tests, Bash, Shell, SaaS,

Achievements

- Led a 6 member team to win the **Smart India Hackathon** 2022 among 3,000 teams, culminating in a presentation to the **Prime Minister of India**. Link
- Led a team of 4 to win the **DivHacks'24** Hackathon hosted by **Columbia University** for developing the project **StrokePlayAR**.
- Won the All India Hackathon HACKNITR with a 4-member team, creating "CommWDeaf", a project focused on improving communication accessibility.